

Auto Ticket Classification using Flow Designer

Project Documentation

1. Introduction

Project Title: Auto Ticket Classification using Flow Designer

Team Members: List team members and their roles.

Name	Role / Phase
Ragu M	Phase 1: Requirement Analysis & Planning
Vanathi R	Team Leader – Phase 2: Backend Development & Configuration
Sanjay J	Phase 3: Automation using Flow Designer & Email Notification
Yoghaprakalya V	Phase 4: Testing, Validation & Security Phase 5: Deployment & Final Documentation

2. Project Overview

Purpose: The school IT helpdesk receives many incident requests daily from students and teachers (Wi-Fi issues, projector failures, password problems, slow computers). IT staff currently review each request manually to identify the issue type, pick a category and subcategory, and inform the caller. This is time-consuming, error-prone and not scalable. This project automates ticket classification at the time of creation using ServiceNow Flow Designer, by analysing keywords in the ticket's short description.

Goals:

- Automatically classify incidents when they are created.
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency.
- Implement a no-code, easily maintainable solution.

Features:

- Automatic assignment of both Category and Subcategory based on keywords in the short description.
- Dependent choice logic between Category and Subcategory.
- Automated email notification to the caller on ticket creation.
- Custom table (Incident WorkFlow) with an auto-numbered ticket ID (INC prefix) for structured, standardised storage.
- Completely no-code, scalable and easy to maintain; packaged as an update set for deployment.

Keyword-to-classification rules:

Keyword in Short Description	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

3. Architecture

This solution is built entirely on the ServiceNow platform and has no custom code. The MERN-style layers are mapped to the ServiceNow platform as follows.

Frontend (User Interface)

The ServiceNow form UI for the Incident WorkFlow table, designed with Form Designer. The student only enters the **Caller** and **Short description**; Category and Subcategory are filled in automatically. Subcategory choices are filtered dynamically by the selected Category.

Backend (Flow Designer Automation)

A flow named **Auto Classify School IT Tickets** (Application: Global) is built in Flow Designer:

- **Trigger:** Record Created on table Incident WorkFlow, condition Category is empty.
- **Flow Logic:** an If / Else If chain that checks whether Short description contains the predefined keywords.
- **Actions:** an Update Record action in each branch sets Category and Subcategory.
- **Email:** a Send Email action sends the confirmation to the caller's email address.

Database (Custom Table and Schema)

Table label **Incident WorkFlow**, name **u_incident_workflow**, Auto-number enabled (Prefix INC, starting number 500, 5 digits). Fields:

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice String	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String Choice	—
Description	Reference	—
State	Reference	New, In progress, On hold, Resolved, Closed
Assigned Group		sys_user_group
Assigned to		sys_user

Dependent choice mapping (Subcategory depends on Category):

Subcategory	Dependent value (Category)
Wi-Fi	network
Projector	hardware
Forgot Password	access
Slow Computer	performance

4. Setup Instructions

Prerequisites

- A ServiceNow instance (e.g., Personal Developer Instance) with **admin** access.
- Flow Designer / Workflow Studio available (Process Automation).
- Outbound email configured on the instance (for notifications).
- Update set XML file exported from the source instance (for importing the solution).
- A modern web browser.

Installation

1 Create an update set: All → Local Update Sets → New. Name: Project Update Set, State: In progress, Application: Global. Save, then click Make This My Current Set.

2 Create the custom table: All → Tables (System Definition) → New. Label: Incident WorkFlow; uncheck Create module; Save. In the Controls tab enable Auto-number and Save again.

3 Create the fields: Open the form → Configure → Form Design and add the fields listed in Section 3. Add choices for Category, Subcategory and State.

4 Configure the dependency: Right-click Subcategory → Configure Dictionary → Advanced view → Dependent Field tab. Enable Use dependent field, set Dependent on field = Category, and add the dependent value for each Subcategory choice.

5 Build the flow: All → Flow Designer → New → Flow. Name: Auto Classify School IT Tickets, Application: Global. Configure the trigger, the If / Else If conditions, the Update Record actions and the Send Email action.

6 Activate the flow using the Activate button at the top right.

7 (Alternative) Import instead of building: On the target instance go to Retrieved Update Sets → Import Update Set from XML, upload the exported file, then Preview and Commit.

There are no environment variables to configure. The only environment dependency is the instance's outbound email setup.

5. Folder Structure

ServiceNow stores configuration as records rather than files. The solution is organised as the following components inside the update set.

Client (Form / UI Components)

Component	Description
Incident WorkFlow form	Default view with Number, Caller, Category, Subcategory, Short description, Description, State, Assigned group, Assigned to.
Form Design layout	Two-column layout created with Form Designer.
Choice lists	Category, Subcategory (dependent) and State choices stored in sys_choice.

Server (Tables and Automation)

Component	Description
Update Set: Project Update Set	Container for all changes; exported as XML (sys_remote_update_set_*.xml).
Table: u_incident_workflow	Custom table with Auto Number (INC prefix) and access role u_incident_workflow_user.
Dictionary entries	Field definitions, including the dependent-field setting on u_subcategory.
Flow: Auto Classify School IT Tickets	Trigger, If / Else If logic, Update Record actions and Send Email action.
System Logs → Emails	Outbound email records used to verify notifications.

6. Running the Application

ServiceNow runs in the cloud, so no local servers need to be started. After activating the flow:

- **Open the application:** navigate to the Incident WorkFlow table (type u_incident_workflow.list in the All menu filter) and click **New**.
- **Create a ticket:** enter the Caller and a Short description, then Submit / Save.
- **Verify:** reload the record; Category and Subcategory are filled in automatically, and an email is generated for the caller.
- **Monitor the flow:** Flow Designer → Executions to see each run of the flow.

7. API Documentation

The solution does not define custom REST endpoints. Because the data lives in a standard ServiceNow table, the platform's built-in **Table API** can be used (the user needs an appropriate role). The Flow Designer trigger fires for records created through the API as well.

Method	Endpoint	Purpose
GET	/api/now/table/u_incident_workflow	List tickets (supports sysparm_query, sysparm_limit)
GET	/api/now/table/u_incident_workflow/{sys_id}	Retrieve a single ticket
POST	/api/now/table/u_incident_workflow	Create a ticket; the flow then sets Category / Subcategory
PUT / PATCH	/api/now/table/u_incident_workflow/{sys_id}	Update a ticket

Method	Endpoint	Purpose
DELETE	/api/now/table/u_incident_workflow/{sys_id}	Delete a ticket

Example request (POST):

```
{ "caller": "<sys_id of user>",
  "short_description": "Projector not turning on" }
```

Example response (after the flow has run, abridged):

```
{ "result": {
  "number": "INC00512",
  "short_description": "Projector not turning on",
  "u_category": "hardware",
  "u_subcategory": "projector" } }
```

Note: the exact column names depend on how the fields were created (custom fields are prefixed with u_).

8. Authentication

- **Authentication:** users sign in to the ServiceNow instance with their platform credentials (session-based login). API access uses basic authentication or OAuth tokens configured on the instance.
- **Authorization:** role-based access control. When the table was created, Create access controls generated ACLs tied to the role **u_incident_workflow_user**. Only users with this role (and administrators) can read, create or modify tickets.
- **Data integrity:** Caller, Assigned group and Assigned to are reference fields that resolve to the sys_user and sys_user_group tables, so only valid users and groups can be selected.

9. User Interface

The ticket form shows the Category and Subcategory fields, with Subcategory options filtered by the chosen Category. The screenshots below show the form design and the dependent-choice behaviour.

Form Design – Category choices

servicenow All Favorites History Workspaces Admin School IT ticket - Create SCH0001002

School IT ticket New record

Number: SCH0001002

Caller: [Search]

Category: Network

Subcategory: -- None --

Short Description: -- None --

Description: Wi-Fi

State: -- None --

Assigned Group: [Search]

Assigned to: [Search]

Submit

Dependent Subcategory – Category: Network shows Wi-Fi

servicenow All Favorites History Workspaces Admin School IT ticket - Create SCH0001002

School IT ticket New record

Number: SCH0001002

Caller: [Search]

Category: Hardware

Subcategory: -- None --

Short Description: -- None --

Description: Projector

State: -- None --

Assigned Group: [Search]

Assigned to: [Search]

Submit

Category: Hardware shows Projector

servicenow All Favorites History Workspaces Admin School IT ticket - Create SCH0001002

School IT ticket New record

Number: SCH0001002

Caller: [Search]

Category: Hardware

Subcategory: Access

Short Description: -- None --

Description: -- None --

Forgot password

State: -- None --

Assigned Group: [Search]

Assigned to: [Search]

Submit

Category: Access shows Forgot Password

servicenow All Favorites History Workspaces Admin School IT ticket - Create SCH0001002

School IT ticket New record

Number: SCH0001002

Caller: [Search]

Category: Performance

Subcategory: -- None --

Short Description: -- None --

Description: Slow Computer

State: -- None --

Assigned Group: [Search]

Assigned to: [Search]

Submit

Category: Performance shows Slow Computer

10. Testing

Strategy: manual, end-to-end functional testing of the automation, plus data-validation and security checks. **Tools:** the Incident WorkFlow form, Flow Designer, and System Logs → Emails (with Preview Email).

Data validation checks

- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the user and group tables.

Test scenarios

#	Input (Short description)	Expected Category	Expected Subcategory	Email	Result
1	Wi-Fi / Network issue (e.g., "My college has issue with Network")	Network	Wi-Fi	Sent to caller	Pass
2	Projector not turning on	Hardware	Projector	Sent to caller	Pass

Procedure: create a new record with a Caller and Short description, save, then reload the form to confirm the fields were filled. To check the email, go to All → Emails (System Logs), search the subject Your Request for the issue has been submitted, open the record and use Preview Email.

11. Screenshots or Demo

The screenshot shows a web application interface for managing 'Project Update Set'. The form includes fields for Name, State, Parent, Release date, Install date, Installed from, and Description. It also displays metadata such as Application (Global), Created (2026-09-29 23:07:08), Created by (admin), and Merged to. Below the form are buttons for Update and Delete, and a section for Related Links. At the bottom, there is a table with columns: Created, Type, View, Target name, Updated by, Remote update set, and Action.

Update set 'Project Update Set' created (Application: Global, State: In progress)

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

* Label: School IT ticket
 * Name: u_school_it_ticket
 Extends table:

Application: Global
 Create module: ☒
 Create mobile module: ☒
 Add module to menu: -- Create new --
 New menu name: School IT ticket
 Remote Table: ☐

Columns Controls Application Access

Accessible from: All application scopes

Can read: ☒
 Can create: ☐
 Can update: ☐
 Can delete: ☐

Allow access to this table via web services: ☒
 Allow configuration: ☒

Submit Cancel Seek

Creating the custom table (System Definition > Tables > New)

Choice List Specification Dependent Field Calculated Value Default Value

Specify a field whose value is used to determine choice list options for this field. [More info](#)

Use dependent field: ☒
 Dependent on field: Category

Create Choice List Delete Column Update

Related Links
[Show Table](#)
[Run Point Scan](#)

Access Controls Choices (4) Attributes (1) Labels (1)

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-09-29 23:35:40	Access
Wi-Fi	wi-fi	en	0	false	2026-09-29 23:35:40	Network
Slow Computer	slow computer	en	3	false	2026-09-29 23:35:41	Performance
Projector	projector	en	1	false	2026-09-29 23:35:40	Hardware

Insert a new row...

Dictionary – dependent field: Subcategory depends on Category, with dependent values

servicenow All Favorites History Workspaces Admin Table - School IT ticket

process automati

FAVORITES
 No Results

ALL RESULTS
 Process Automation
 Getting Started
 Workflow Studio
 Flow Designer
 Triggers
 FDH Dashboard
 Playbook Designer
 Flow Administration
 Today's Executions
 Active Flows
 Content Definitions
 Content Filtering Rules

Related Links
[Form Builder](#)
[Design Form](#)
[Layout List](#)
[Show Form](#)
[Show List](#)
[Show Schema Map](#)
[Add to Service Catalog](#)
[Run Point Scan](#)

the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More info](#)

Label: School IT ticket
 Name: u_school_it_ticket
 Application: Global
 Remote Table: ☐

Access

Accessible from: All application scopes

Can read: ☒
 Can create: ☐
 Can update: ☐
 Can delete: ☐

Allow access to this table via web services: ☒
 Allow configuration: ☒

Opening Flow Designer from the Process Automation menu

Create SCH0001006 | School IT ticket | ServiceNow - Profile 2 - Microsoft Edge

https://dev343140.service-now.com/u_school_it_ticket.do?sysparm_dear_stack=true&sysparm_reference=u_school_it_ticket...

School IT ticket

New record

Number: SCH0001006

State: New

Caller: Abel Tuter

Assigned Group:

Category: Network

Assigned to:

Subcategory: Wi-Fi

Short Description: Network issue

Description:

Submit

Test scenario 1 – ticket created with short description "Network issue" (Network / Wi-Fi)

servicenow

AllFavoritesHistoryWorkspacesAdminEmails

Created

Search

Actions on selected rows...New

Created	Recipients	Subject	Type	Notification type	User ID
2026-10-01 00:10:31	abel.tuter@example.com	Your Request for the issue has been submitted	send-ready	SMTP	(empty)
2026-10-01 01:00:06	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-10-01 01:00:07	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)
2026-10-01 00:02:30	abel.tuter@example.com	Your Request for the issue has been submitted	send-ready	SMTP	(empty)
2026-10-01 00:02:56	abel.tuter@example.com	Your Request for the issue has been submitted	send-ready	SMTP	(empty)
2026-10-01 00:00:15	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
2026-10-01 01:00:05	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-10-01 00:11:07	abel.tuter@example.com	Your Request for the issue has been submitted	send-ready	SMTP	(empty)

Email log – confirmation emails to callers

[illegible]

Preview Email – confirmation message

Update set exported as XML for deployment

12. Known Issues

- **Keyword matching is literal:** the conditions use contains on exact text (e.g., "Wi-Fi", "Network", "Forgot password", "Slow computer"). Variants such as "WiFi", "wifi" or misspellings may not match, so the ticket stays unclassified.
- **Only the first matching rule applies** (If / Else If chain); a ticket mentioning several issues gets one classification.
- **Keywords listed in the requirement** (e.g., Login, Slow, Hanging) are broader than the conditions currently configured (e.g., "Forgot password", "Slow computer").
- **Email delivery depends on the instance's outbound email configuration;** on a developer instance emails may only appear as send-ready in the Emails log.
- **Subcategory spelling:** keep choice labels consistent (e.g., "Slow Computer") to avoid mismatches in flow values.

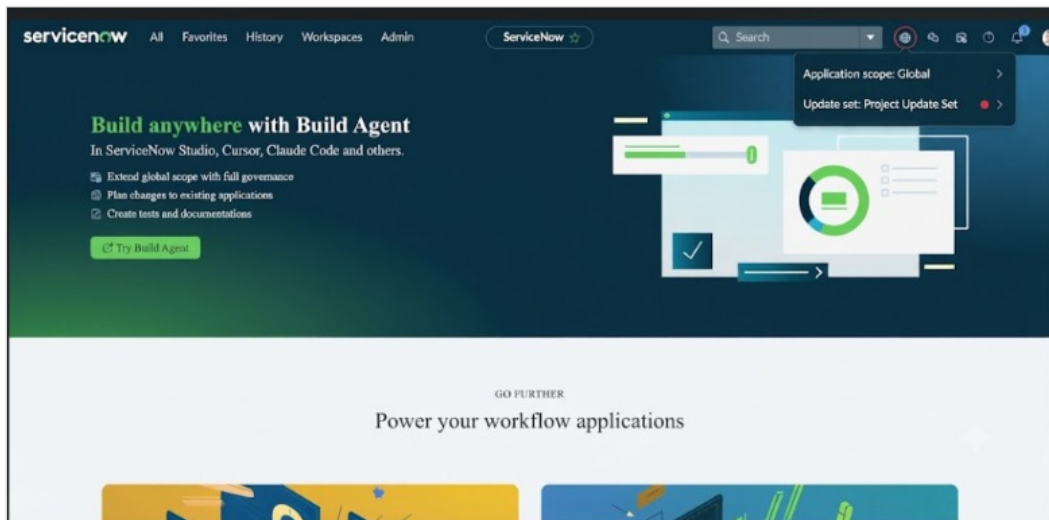
13. Future Enhancements

- Assignment automation: automatically set Assigned group / Assigned to based on the category.
- SLA tracking and escalation for unresolved tickets.
- Predictive Intelligence to classify tickets using machine learning.
- Expand keyword coverage (synonyms, variants, case handling) and add more categories, ideally using a decision table or a lookup so rules are easier to maintain.
- Add a Service Portal / catalog form for students, and notifications on resolution or state changes.

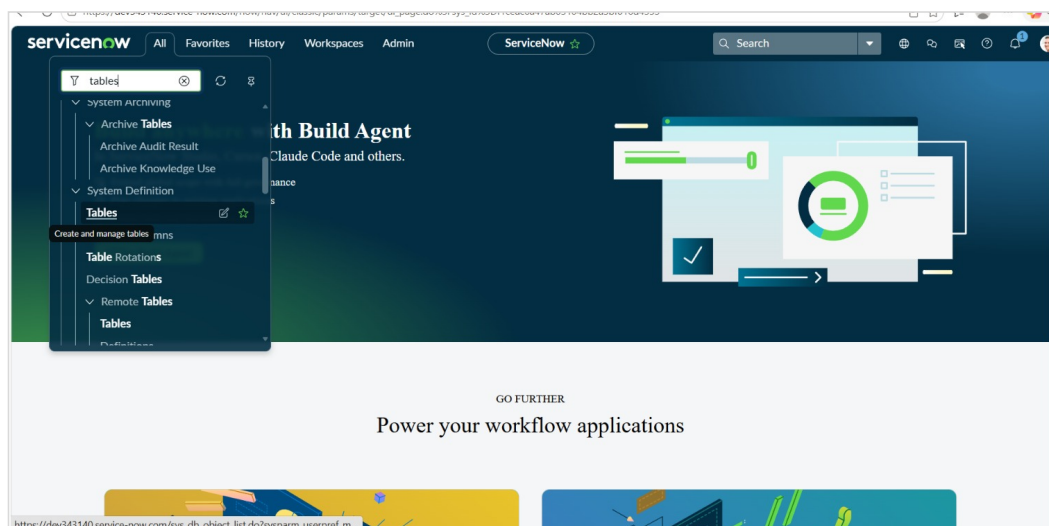
Conclusion

The project delivers an end-to-end, no-code automation for the school IT helpdesk. Tickets are classified automatically on creation, dependent choice fields keep data accurate, and callers are notified immediately by email. The design suits small to medium environments such as educational institutions and can be extended with assignment automation, SLA tracking or predictive intelligence.

Appendix – Additional Screenshots



Update set set as current (Application scope: Global, Update set: Project Update Set)



Opening Tables (System Definition) from the All menu

The screenshot shows the 'Tables' list in the ServiceNow System Definition. The table has columns: Label, Name, Extends table, Extensible, and Updated. The data is as follows:

Label	Name	Extends table	Extensible	Updated
Account Subscription Entitlement	account_subscription_entitlement	(empty)	false	2026-06-12 14:52:23
Adaptive Authentication Event	adaptive_auth_event	(empty)	false	2026-06-12 14:41:06
Agent Assist Recommendation	agent_assist_recommendation	Application File	false	2026-06-12 14:35:51
AI/ML Server File	agent_file	(empty)	false	2026-06-12 14:22:34
AI/ML Access Policy	alg_access_policy	Application File	false	2026-06-12 14:01:27
AI/ML Access Policy Scope Mapping	alg_access_policy_scope_mapping	Application File	false	2026-06-12 14:01:28
AI/ML Scope Tool Mapping	alg_scope_tool_mapping	Application File	false	2026-06-12 14:01:26
Record Producer Configuration	ais_rp_config	Application File	false	2026-06-12 14:39:11
Search Actions	aisa_ui_action	Application File	false	2026-06-12 14:39:11
AI Search ACL Overrides	ais_acl_overrides	Application File	false	2026-06-12 14:03:37
AI Search Active Table Ingestion Tracker	ais_active_table_ingestion_tracker	(empty)	false	2026-06-12 14:03:35
AI Search Transaction Log	ais_api_transaction_log	(empty)	false	2026-06-12 14:03:39
AI Search Async Genius Result	ais_async_genius_result	(empty)	false	2026-06-12 14:03:39
AI Search Async Request	ais_async_request	(empty)	false	2026-06-12 14:03:40
AI Search Child Table	ais_child_table	Application File	false	2026-06-12 14:03:38
AI Search Configuration Attribute	ais_configuration_attribute	(empty)	false	2026-06-12 14:03:35
AI Search Connection	ais_connection	(empty)	false	2026-06-12 14:03:39
AI Search Country To Search Language	ais_country_to_search_language	Application File	false	2026-06-12 14:03:39
Custom Matcher	ais_custom_matcher	Application File	false	2026-06-12 14:03:37
AI Search Indexed Source	ais_datasource	Application File	false	2026-06-12 14:03:38

Tables list in System Definition

Appendix – Additional Screenshots (continued)

The screenshot shows the 'Table Controls' tab in a software interface. It features a 'Columns' tab, a 'Controls' tab, and an 'Application Access' tab. Under the 'Controls' tab, there are checkboxes for 'Extensible' and 'Live feed', both of which are unchecked. A blue informational banner states: 'Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format'. Below this, the 'Auto-number' checkbox is checked. There are three input fields: 'Prefix' with the value 'INC', 'Number' with the value '1000', and 'Number of digits' with the value '7'.

Table Controls tab – Auto-number enabled (Prefix INC)

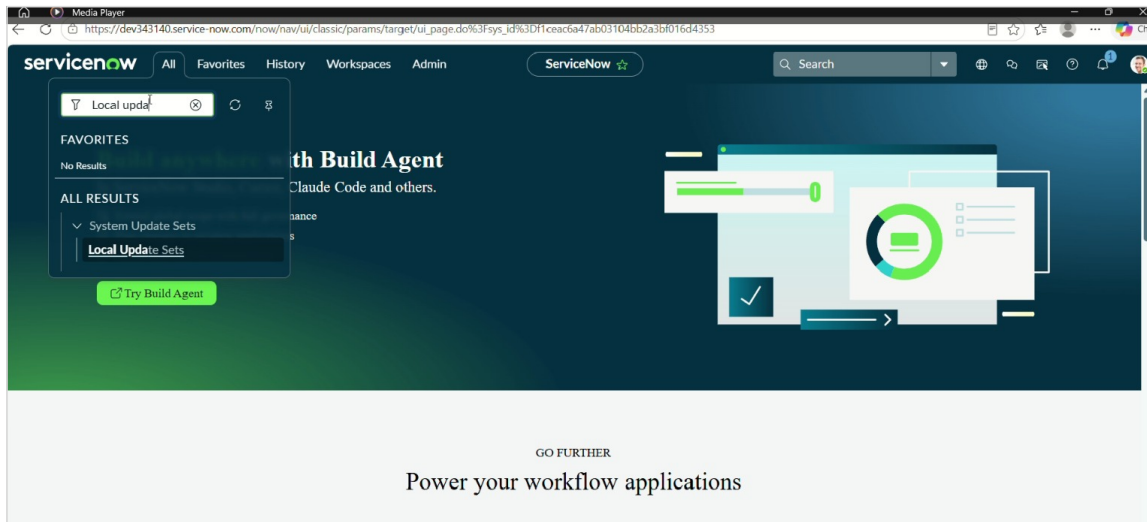
The screenshot shows the 'Form Design' interface for a 'School IT ticket' form. A 'Properties' dialog box is open, showing the 'Choices' section. The 'Choice type' is set to 'Dropdown with none'. Below this, there is a list of choices with their corresponding values and a 'Read only' status. The choices are: 'Wi-Fi' (value: wi-fi), 'Projector' (value: projector), 'Forgot Password' (value: forgot password), and 'Slow Computer' (value: slow computer). Each choice has a blue dot indicating it is selected or active. The background shows the 'Form Design' workspace with a 'School IT ticket' form and a 'Fields' list on the left.

Form Design – Subcategory choices (Wi-Fi, Projector, Forgot Password, Slow Computer)

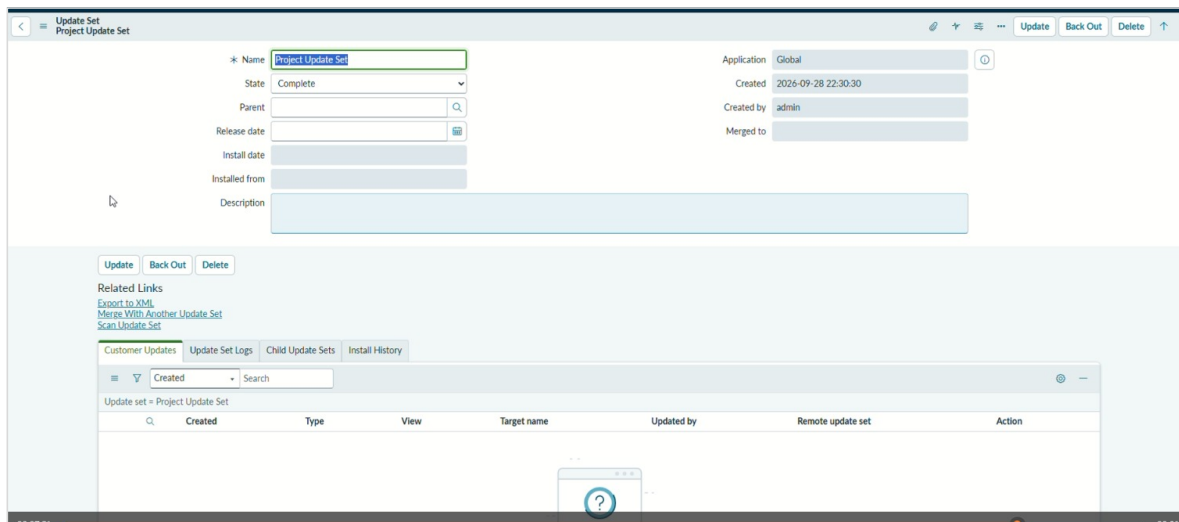
The screenshot shows the 'Dependent Field' tab in a software interface. It features a 'Choice List Specification' tab, a 'Dependent Field' tab, a 'Calculated Value' tab, and a 'Default Value' tab. Under the 'Dependent Field' tab, there is a blue informational banner that states: 'Alters the behavior of a field or functionality that depends on the field. [More Info](#)'. Below this, there is an 'Attributes' section with the value 'edge_encryption_enabled=true'. At the bottom, there is a 'Use dependent field' checkbox which is checked, and a 'Dependent on field' dropdown menu with the value 'Category'.

Dependent Field tab – Use dependent field enabled, Dependent on field = Category

Appendix – Additional Screenshots (continued)



Opening Local Update Sets from the All menu



Update set 'Project Update Set' marked Complete